

Aleksandr Gordeev

Izmir, Türkiye | algordeev18@gmail.com | [LinkedIn](#) | [GitHub](#)

EDUCATION

Yaşar University

B.S. in Computer Engineering, 100% Scholarship

Izmir, Türkiye

2023 – Expected 2027

Universidad Complutense de Madrid

Erasmus+ Exchange, Grant Recipient

Madrid, Spain

2024 – 2025

EXPERIENCE

Robotics and Autonomous Systems Intern

ISCAR Research Group, Universidad Complutense de Madrid

Jun 2026 – Jul 2026

Madrid, Spain

- Extended a MATLAB trajectory-planning framework from a single vehicle to coordinated multi-ASV missions for cyanobacterial-bloom monitoring.
- Implemented and evaluated NSGA-II and NSGA-III optimization with collision avoidance, route-overlap reduction, and workload-balancing objectives.
- Ran parallel simulation campaigns with environmental data and analyzed Pareto fronts using non-parametric statistical methods.

Assistant Robotics Instructor and Competition Judge

Palace of Children's Creativity

2018 – 2022

Nizhny Novgorod, Russia

- Guided junior students in robotics projects and evaluated mechanics and algorithms at youth competitions.

Robotics Mentor and Event Coordinator

Regional Charity Organisation "Zabota"

2018 – 2020

Nizhny Novgorod, Russia

- Taught LEGO EV3 and Arduino fundamentals to underprivileged youth and helped organize robotics events.

PROJECTS

Red Monopoly | Java, OOP, Networking, GUI, Scrum | [GitHub](#)

Jan 2025 – May 2025

- Developed a networked desktop game supporting IP-based multiplayer for up to eight players.
- Applied object-oriented design patterns to implement adaptive AI and game status functionality while separating game logic from the GUI.

Autonomous Tram System | C++, Arduino, PCB Design | [GitHub](#)

2020 – 2022

- Built an autonomous model tram that won the WRO 2021 Russian National Final and competed at the international final.
- Developed modular real-time firmware for motion control, obstacle detection, fail-safe behavior, and vehicle-to-infrastructure communication.
- Designed custom PCBs integrating an Arduino Nano, motor drivers, wireless modules, distance sensors, and actuators.

SELECTED ACHIEVEMENTS

International Finalist – World Robot Olympiad (WRO) 2021

First Place – All-Russian Festival of Research Design Works "Vector" (2022)

Winner (Ecological Urban Infrastructure) – Russian Robot Olympiad 2021

Second Place – Baltic Science and Engineering Competition (2022)

Third Place – XIV All-Russian Youth Competition ROST-ISEF (2022)

SKILLS

Programming: C/C++, MATLAB, Java, SQL, Python

Robotics: Autonomous Systems, Trajectory Planning, Multi-Robot Systems, Collision Avoidance

Low-level: MIPS, RISC-V Assembly

Engineering: Arduino, Embedded Systems, PCB Design, Digital Design, Electronics

CS Concepts: OOP, Data Structures, Algorithms, Computer Networks, Scrum

Tools: Git, GitHub, LaTeX

Languages: Russian (Native), English (C1), Turkish (B1), Spanish (A2)